

PROJECT PROPOSAL.

Group ID:
J26-ISE-313

Specialization:
Information Systems Engineering

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PROBLEM INTRODUCTION VIDEO

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USERS

**Technology Startup
Founders/Entrepreneurs**

**Technology Company Product
Managers**

**Technology Company
Business Leaders / Managers**

STAKEHOLDERS

**Sri Lankan Technology
Companies**

**Infrastructure / Cloud /
DevOps Professionals**

**Internet Service Providers
(ISPs)**

**Sri Lankan Technology
Companies**

**Legal & Compliance
Professionals**

**Digital Marketing
Professionals**

**Potential Product Users /
Customers**

**Startup Ecosystem
Organizations / Incubators**

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EXISTING SOLUTIONS AND THEIR LIMITATIONS

Existing solutions provide support for individual business areas, but they do not provide an integrated, automated and Sri Lanka-specific assessment of technology product readiness across marketing, regulations, product/service management and infrastructure.

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THE PROPOSED SOLUTION

A Smart Consulting Platform for Sri Lankan Technology Product Success that evaluates Marketing, Rules & Regulations, Product & Service Management, and Infrastructure to identify risks and provide customized recommendations before and during product development.

... **INFRASTRUCTURE ASSESSMENT COMPONENT**



Contribution of my component to the overall solution:

This component helps prevent startup failure by ensuring companies have adequate infrastructure to operate and scale their technology products, while identifying infrastructure gaps before they lead to disruptions, higher costs, or growth limitations.

Novelty of My Component:

An automated Infrastructure Readiness and Scalability Advisor that analyses company infrastructure requirements, identifies infrastructure gaps, provides a readiness score, and generates a customized scalability plan for future growth.

... **INFRASTRUCTURE ASSESSMENT COMPONENT**

Problem Definition:

Existing infrastructure assessment methods do not provide a Sri Lanka-specific and automated approach for identifying infrastructure gaps and evaluating the readiness of technology startups to operate and scale successfully.

Research Objective:

To address this gap, this research proposes an automated infrastructure readiness solution that assesses a company's infrastructure, identifies infrastructure gaps, and generates a customized plan for infrastructure improvements and future scalability.

... **INFRASTRUCTURE ASSESSMENT COMPONENT**

Subjectives:

1. Identify infrastructure challenges
2. Collect real company infrastructure data
3. Identify available infrastructure options
4. Define infrastructure readiness criteria
5. Design data-collection forms
6. Build the Infrastructure Readiness Checklist
7. Build the Scalability Planning Module
8. Generate the final assessment report
9. Test, validate, and improve the system

... INFRASTRUCTURE ASSESSMENT COMPONENT

Application of Domain Knowledge:

Domain: Infrastructure Assessment, Decision Support, and Scalability Planning.

Technologies: Full-stack web development, database systems, rule-based decision-making, and scoring models, RAG Architecture.

Validation & Evaluation: Evaluate the readiness score and recommendations with selected Sri Lankan tech startup companies and infrastructure specialists and compare the results with expert assessments.

Data Availability: Primary data from surveys/interviews; secondary data from ISP, cloud-provider and infrastructure information, research papers.

Security & Ethics: Obtain ethical approval, collect informed consent, anonymize participant responses and securely store research data.

... **INFRASTRUCTURE ASSESSMENT COMPONENT**

Implementation Details:

System Overview:

Company Context → Infrastructure Needs → Relevant Questions → Specialized Infrastructure Agent → Readiness & Gap Analysis → Scalability Plan → RAG Report & Dashboard

Specialized Agents:

- **Analyses infrastructure responses**
- **Identifies infrastructure gaps**
- **Supports readiness assessment**
- **Provides inputs for scalability recommendations**

... **INFRASTRUCTURE ASSESSMENT COMPONENT**

Implementation Details:

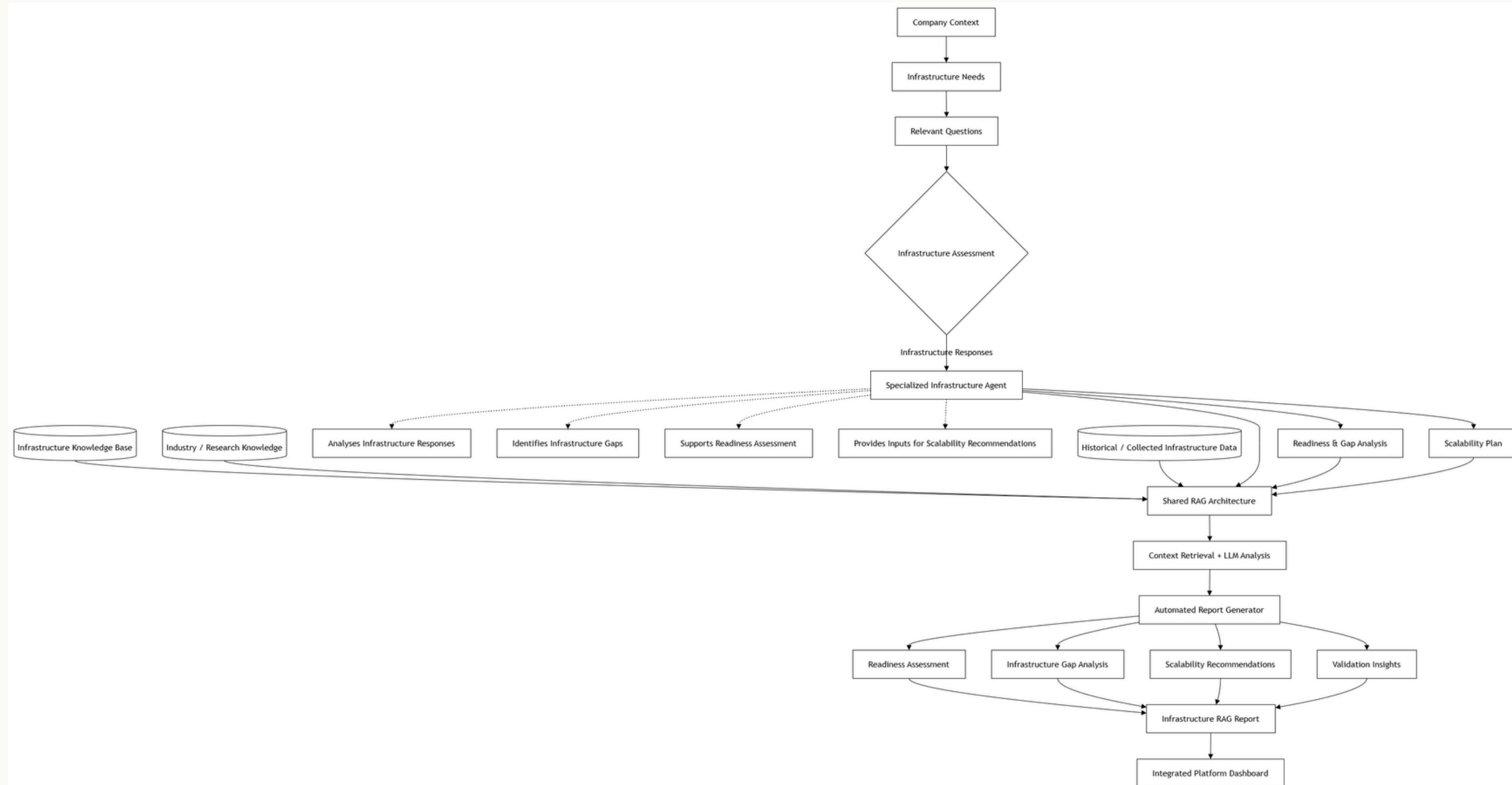
Dependencies:

Questionnaire Data • Rule-Based Scoring Engine • Infrastructure Knowledge Base • RAG / LLM • Shared Database

Common Components:

User Profile • API Layer components • Report Generator • Integrated Dashboard • Shared Database

... INFRASTRUCTURE ASSESSMENT COMPONENT



... **PRODUCT PERFORMANCE ASSESSMENT**



Contribution of my component to the overall solution:

This component reduces the risk of startup failure by ensuring products are built around real customer needs and market demand, while identifying weak features and market barriers before valuable time and money are wasted.

Novelty of My Component:

An automated Product and Service Validation system that turns questionnaire responses into one customized report combining SWOT analysis, market research guidance, and potential market blocker identification to support better product decisions.

... **PRODUCT PERFORMANCE ASSESSMENT**

Problem Definition:

Current product validation methods are mostly generic and manual, leaving Sri Lankan technology startups without an automated way to turn customer and market information into clear evidence about product fit, weaknesses, and possible market barriers before making costly product decisions.

2) **Market research.** For start-ups, market research is a vital success factor. It is important to find a market niche for a new innovative product and be able to fill this niche. Seven participants (58%) reported performing market research to evaluate market requirements, but only after the organization raised funds, recruited workers, and conducted a feasibility study to assess the applicability of scientific and technical concepts. The production of the company's products had little to no gain from doing experiments at a later point since the money had already been raised and workers had already been recruited to produce a particular product. At this point, the pivot was practically and literally impossible, leading to start-up failure. The absence of timely market research was a cause of start-up failure, six participants (50%) suggested. These results are comparable to previous research (Song et al., 2008). As such, conducting timely market research may result in higher technology start-up success rates.

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... **PRODUCT PERFORMANCE ASSESSMENT**

Research Objective:

To address this gap, this research proposes :

An automated product and service validation solution that analyses questionnaire responses and generates a customized report combining SWOT analysis, market research guidance, and potential market blockers to support better product decisions

... **PRODUCT PERFORMANCE ASSESSMENT**

Subjectives:

1. Review product validation and market research methods
2. Design questionnaires for different product stages
3. Build the RAG-based response analysis module
4. Generate an automated SWOT analysis
5. Generate market research guidance
6. Identify potential market blockers
7. Test, validate, and improve the system

... **PRODUCT PERFORMANCE ASSESSMENT**

Application of Domain Knowledge:

Domain: Product & Service Management, Product Validation, Market Research, and Customer Needs Analysis.

Technologies: Multi-Agent Architecture, RAG, Gemini 2.5 Flash , Gemini Embedding , Pinecone Vector DB , Supabase , Full-Stack Web Application

Validation & Evaluation: Test the solution with selected startups, product managers, and users, and improve the generated results based on their feedback.

Data Availability: Primary data from surveys, interviews, and questionnaires; secondary data from research papers, case studies, and market reports.

Security & Ethics: Obtain participant consent, anonymize responses, securely store collected data, and follow ethical approval requirements.

... **PRODUCT AND SERVICE MANAGEMENT**

Implementation Details:

System Overview

**Questionnaire → Product Stage Detection → Specialized Agent → RAG Analysis →
Validation Report → Dashboard**

Specialized Agents:

- **Proposed Product Agent**
- **Product Improvement Agent**
- **Failed Product Analysis Agent**

... **PRODUCT AND SERVICE MANAGEMENT**

Implementation Details:

Dependencies:

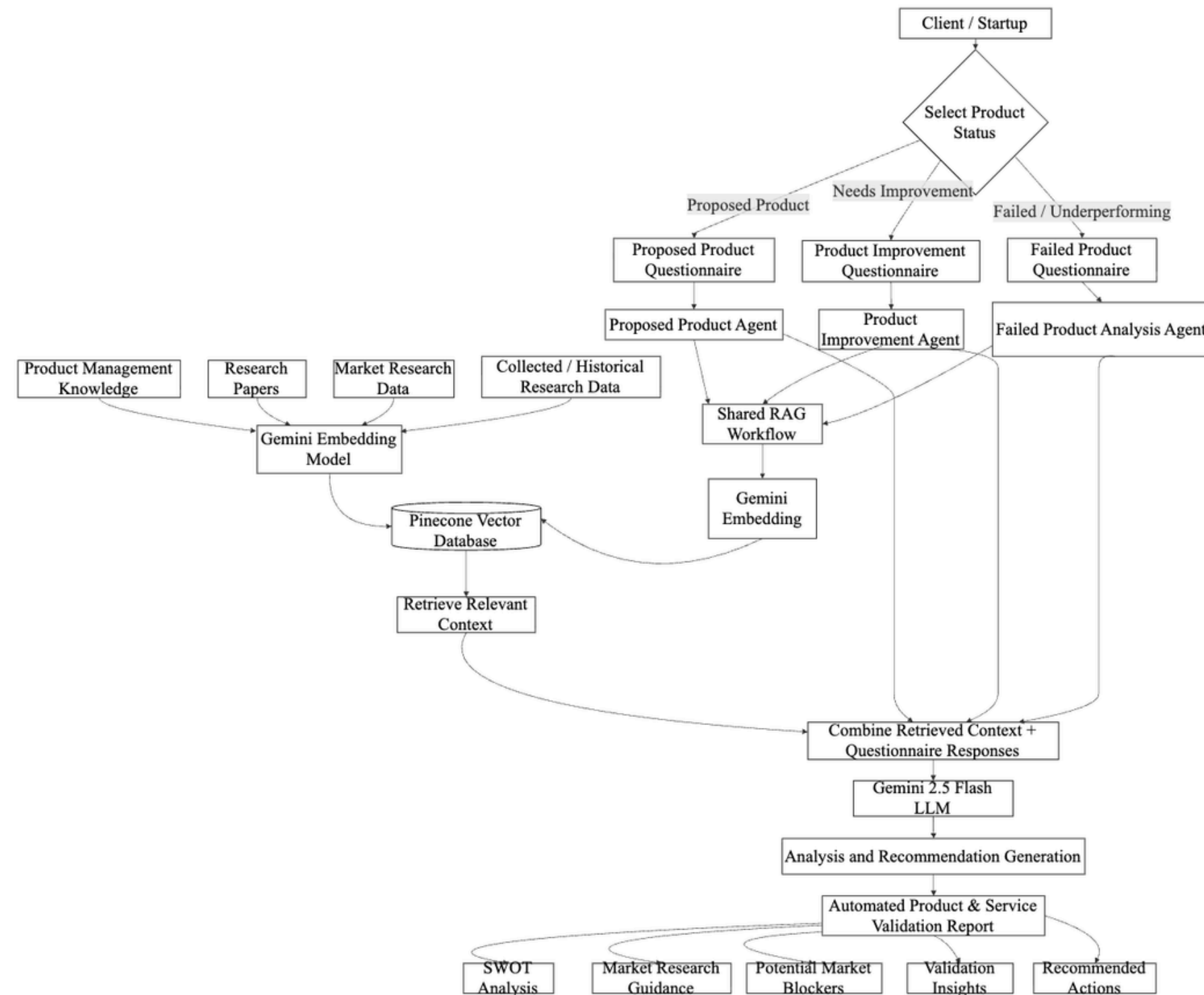
RAG/LLM • Knowledge Base • Vector DB • Questionnaire Data • Database

Common Components:

**User Profile • Shared Database • API Layer • Report Generator • Integrated
Dashboard**

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PRODUCT AND SERVICE MANAGEMENT



... **MARKETING COMPONENT**



Contribution of my component to the overall solution:

Helps tech companies communicate product value clearly, find competitor gaps, and select the right channels to reach customers successfully.

Novelty of My Component:

- **Audience Technical Literacy Classifier**
- **Smart Channel & Budget Recommendation System**

... **MARKETING COMPONENT**

Problem Definition:

Sri Lankan tech products often fail post-launch because companies use jargon-heavy technical language with non-technical customers and lack data-driven channel or budget strategies tailored to the local market.

Research Objective:

To develop an AI-based tool that simplifies technical marketing messages and recommends suitable marketing channels and budgets for Sri Lankan tech companies.

... **MARKETING COMPONENT**

Subjectives:

1. Analyze Sri Lankan digital marketing frameworks and market data.
2. Collect raw user inputs including product text, target audience demographics, monthly budget (LKR), and conversion goals.
3. Classify target audiences based on technical literacy using NLP.
4. Translate technical features into simple customer-value messages.
5. Rank suitable marketing channels using the AHP algorithm.
6. Develop an automated budget allocation system.
7. Generate target audience profiles, campaign execution plans, and summary reports.

... **MARKETING COMPONENT**

Application of Domain Knowledge:

Domain: Digital Marketing, Marketing Analytics, Natural Language Processing (NLP), Decision Science

Technologies: NLP & Text Processing, AI / Machine Learning, AHP Algorithm, Gemini 2.5 Flash API, Supabase (pgvector for RAG), Analytic Hierarchy Process (AHPy)

Validation & Evaluation: Surveys and interviews with Sri Lankan startup founders, Evaluate audience classification and channel recommendations, Compare recommendations with expert/industry opinions

Data Availability: Product descriptions, Target demographics, Marketing budgets in LKR, Conversion goals & target regions, Sri Lankan digital market metrics

Security & Ethics: Protect participant and company information, Obtain informed consent for surveys/interviews, Maintain confidentiality and responsible use of AI-generated recommendations

... **MARKETING COMPONENT**

Implementation Details:

System overview

Intake Form → Technical Literacy Scoring → Multi-Agent Processing → AHP Channel Ranking & Budget Optimization → Marketing Strategy Dashboard

Specialized Agents

- **Audience Intelligence Agent**
- **Feature-to-Value Storytelling Agent**
- **Channel Ranking Agent**
- **Budget Optimization Agent**

... **MARKETING COMPONENT**

Implementation Details:

Dependencies:

- NLP/LLM • Local Market Knowledge Base • Supabase (pgvector) • Questionnaire Data • AHP Engine (AHPy)

Common Component:

- User Profile • Shared Database • API Layer (FastAPI) • Report Generator • Integrated Dashboard

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MARKETING COMPONENT



... DULANJANA'S COMPONENT



Contribution of my component to the overall solution:

Build a tool that helps Sri Lankan tech companies easily find and follow legal, tax, and data laws before and after launching products.

Novelty of My Component:

A searchable Sri Lanka regulatory database combined with a pre-build checker to identify legal risks before product development.

... **DULANJANA'S COMPONENT**

Problem Definition:

Sri Lankan tech companies struggle to find and navigate complex legal, tax, and data regulations, leading to compliance risks before and after launch.

Research Objective:

To build a tool that helps Sri Lankan tech companies easily identify regulations and evaluate pre-build compliance risks for their products.

... **DULANJANA'S COMPONENT**

Subjectives:

1. Review Sri Lankan legal, tax, and data protection laws for tech products.
2. Interview legal, compliance, and industry professionals.
3. Collect and organize relevant regulatory information into a single dataset.
4. Develop the searchable Sri Lanka Regulatory Mapping Module.
5. Build the Pre-Build Compliance Checker to detect legal risks before development.

... DULANJANA'S COMPONENT

Application of Domain Knowledge:

Domain:

Technologies: LLM: Google Gemini API, Vector DB: Supabase , Backend: Python , Framework: RAG Architecture

Validation & Evaluation:

Data Availability: Primary data from interviews with legal, compliance, and industry professionals; secondary data from Sri Lankan regulatory documents, acts (PDPA, Inland Revenue Act)

Security & Ethics: Implement strict data privacy protocols by masking user product details before processing, ensuring no proprietary business ideas are stored.

... **DULANJANA'S COMPONENT**

Implementation Details:

System Overview :

Product Idea Intake Form → Compliance Risk Detection → Specialized Agent Analysis → RAG Workflow (Supabase + Gemini) → Compliance Report → Dashboard

Specialized Agents:

- **Regulatory Mapping Agent**
- **Pre-Build Compliance Agent**

... **DULANJANA'S COMPONENT**

Implementation Details:

Dependencies:

Gemini LLM
RAG Architecture
Supabase
Regulatory Knowledge Base
User Intake Form Data

Common Components:

User Profile • Supabase Database • API Layer • Report Generator • Integrated Dashboard

... **WORK BREAKDOWN STRUCTURE**

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filter=&groupBy=none](https://amashirajapaksha622.atlassian.net/jira/software/projects/SCRUM/boards/1?filter=&groupBy=none)

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GANTT CHART

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ABILITY OF COMMERCIALIZATION